MoonWalker Weekly Meeting

10/1 - 10/7 Summary

Elijah, Rowel, Martin

Progress Update

- Elijah
  - Installed Omniverse Isaac Sim in DT Server
    - Created documentation for installing and testing isaac sim
    - Loaded in files from Dr. Bingbing Li (Jared's files)
  - Explored movement scripts for Isaac Sim
    - https://docs.isaacsim.omniverse.nvidia.com/latest/robot_simulation/mobile_robot_controllers.html?utm_source=chatgpt.com
    - https://docs.isaacsim.omniverse.nvidia.com/latest/py/source/extensions/isaacsim.robot_motion.motion_generation/docs/index.html
- Martin
  - Installed Unity Hub on new laptop
  - Resolved compilation issue caused by outdated/old kafka scripts files in the Unity HSML project; In-progress work with unityBL
  - Initial unityServerTest set up
  - Document steps for unity_hsml (Currently undocumented on github, had to figure out steps on my own)
- Rowel
  - Set up MySQL Database in DT Server
  - Set up the HSML_API on the DT Server
  - Connected HSML_API with Kafka and MySQL

Challenges

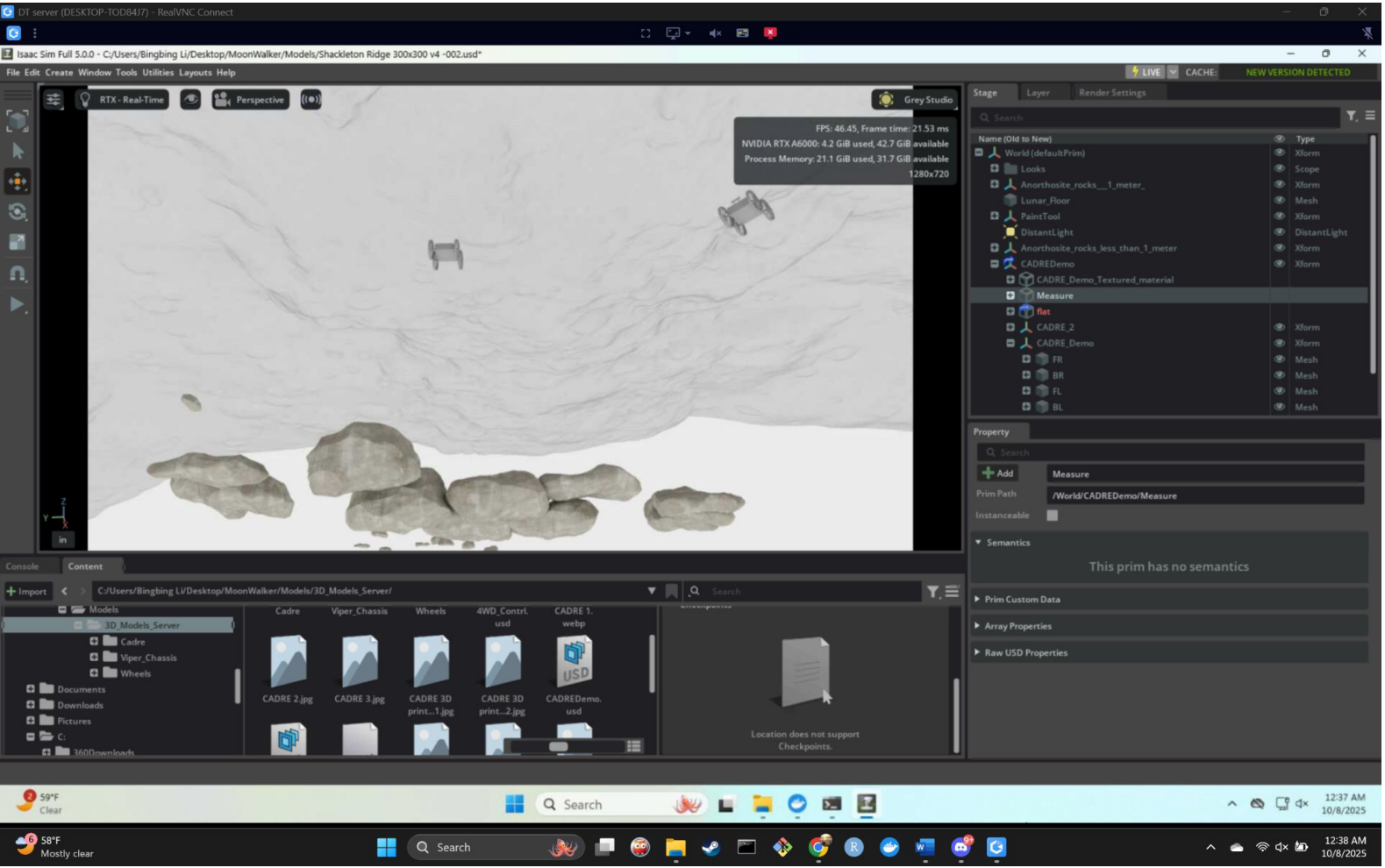
- Resolving dependency conflicts caused by multiple requirements.txt files in the HSML_API project, which were breaking Docker builds and leading to inconsistent environments.
- Configuring stable connectivity between HSML_API, Kafka, and MySQL within a WSL2 setup, overcoming network isolation issues between Docker containers and and WSL2
- There’s an issue with git lfs for Unity HSML, some files can’t be pulled that are needed because git lfs is no longer enabled for the repo.

Missing Dependencies Caused by Conflicting requirements.txt files

```
Ubuntu
  File "/usr/local/lib/python3.10/asyncio/runners.py", line 44, in run
    return loop.run_until_complete(main)
  File "/usr/local/lib/python3.10/asyncio/base_events.py", line 649, in run_until_complete
    return future.result()
  File "/usr/local/lib/python3.10/site-packages/uvicorn/server.py", line 70, in serve
    await self._serve(sockets)
  File "/usr/local/lib/python3.10/site-packages/uvicorn/server.py", line 77, in _serve
    config.load()
  File "/usr/local/lib/python3.10/site-packages/uvicorn/config.py", line 435, in load
    self.loaded_app = import_from_string(self.app)
  File "/usr/local/lib/python3.10/site-packages/uvicorn/importer.py", line 22, in import_from_string
    raise exc from None
  File "/usr/local/lib/python3.10/site-packages/uvicorn/importer.py", line 19, in import_from_string
    module = importlib.import_module(module_str)
  File "/usr/local/lib/python3.10/importlib/__init__.py", line 126, in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
  File "<frozen importlib._bootstrap>", line 1050, in _gcd_import
  File "<frozen importlib._bootstrap>", line 1027, in _find_and_load
  File "<frozen importlib._bootstrap>", line 1006, in _find_and_load_unlocked
  File "<frozen importlib._bootstrap>", line 688, in _load_unlocked
  File "<frozen importlib._bootstrap_external>", line 883, in exec_module
  File "<frozen importlib._bootstrap>", line 241, in _call_with_frames_removed
  File "/app/src/main.py", line 3, in <module>
    from src.registration import router as registration_router # Import the router from registration_api
  File "/app/src/registration.py", line 7, in <module>
    from cryptographic_tool import extract_did_from_private_key
ModuleNotFoundError: No module named 'cryptographic_tool'
moonwalker@DESKTOP-TQDB4J7:~/hsml_api$ #
moonwalker@DESKTOP-TQDB4J7:~/hsml_api$
```

HSML_API Failing to connect to Kafka Server

```
Ubuntu   Ubuntu
essed)
%3|1759911339.498|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911369.490|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911369.507|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911399.500|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911399.516|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911429.508|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911429.526|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911459.518|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
%3|1759911459.535|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: localhost:9092/bootstrap: Connec
t to ipv4#127.0.0.1:9092 failed: Connection refused (after 0ms in state CONNECT, 30 identical error(s) suppr
essed)
^[#
```



Ridge and Cadre USD files running on DT Server

HSML_API successfully connected to Kafka server

```
moonwalker@DESKTOP-TOD84J7:~/hsml_api$ docker run -d -p 8000:8000 -e KAFKA_BOOTSTRAP=172.26.30.154:9092 hsml
_api
a66f6e20d480287c0da470f1d92bf3aa7f711f174c9b4094a6c4cd59915f03d9
moonwalker@DESKTOP-TOD84J7:~/hsml_api$ docker exec -it $(docker ps -q --filter ancestor=hsml_api) bash -lc \

'python -c "from confluent_kafka import Producer; p=Producer({\"bootstrap.servers\":\"$KAFKA_BOOTSTRAP\"});
print(p.list_topics(timeout=5).brokers)"'
{1: BrokerMetadata(1, localhost:9092)}
moonwalker@DESKTOP-TOD84J7:~/hsml_api$
```

Database successfully connected to HSML_API

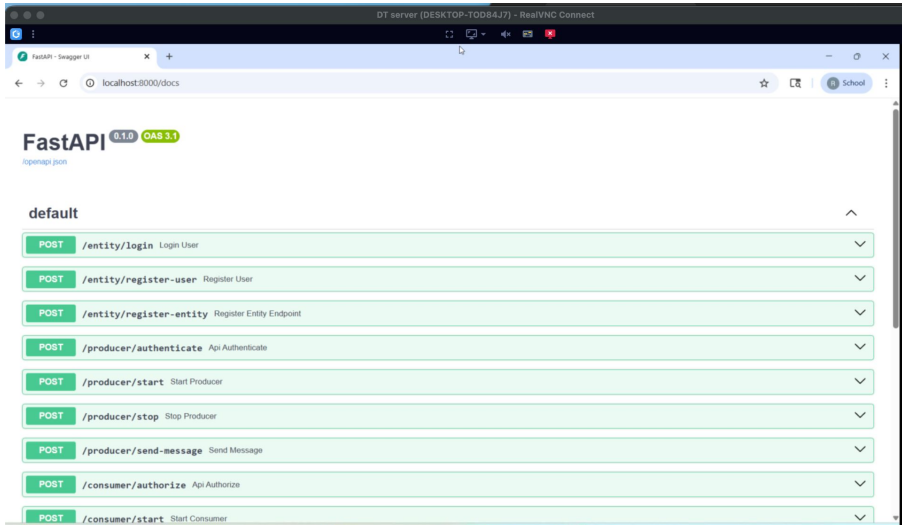
```
Ubuntu   Ubuntu   Ubuntu

ne 365, in _open_connection
    self._cmysql.connect(**cnx_kwargs)
_mysql_connector.MySQLInterfaceError: Host 'DESKTOP-TOD84J7.mshome.net' is not allowed
to connect to this MySQL server

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "<string>", line 1, in <module>
  File "/usr/local/lib/python3.10/site-packages/mysql/connector/pooling.py", line 322,
in connect
    return CMySQLConnection(*args, **kwargs)
  File "/usr/local/lib/python3.10/site-packages/mysql/connector/connection_cext.py", li
ne 153, in __init__
    self.connect(**kwargs)
  File "/usr/local/lib/python3.10/site-packages/mysql/connector/abstracts.py", line 152
9, in connect
    self._open_connection()
  File "/usr/local/lib/python3.10/site-packages/mysql/connector/connection_cext.py", li
ne 370, in _open_connection
    raise get_mysql_exception(
mysql.connector.errors.DatabaseError: 1130 (HY000): Host 'DESKTOP-TOD84J7.mshome.net' i
s not allowed to connect to this MySQL server
moonwalker@DESKTOP-TOD84J7:~/hsml_api$ docker exec -it $(docker ps -q --filter ancestor
=hsml_api) bash -lc \
"python -c \"import mysql.connector as mc; print(mc.connect(host='172.26.30.154', user=
'did_app', password='csun2014', database='did_registry').is_connected())\""
True
moonwalker@DESKTOP-TOD84J7:~/hsml_api$
```

FastAPI Swagger UI on DT Server



Action Plan

- Test API endpoints to verify functionality, data accuracy, and reliability.
- Integrate the HSML Web Form Application with the API backend, ensuring seamless data flow between the user interface and the HSML_API services.
- Begin testing with Omniverse, then extend evaluation to the Unity environment.
    - Construct path planning scripts
- Continue research by exploring new techniques and optimization methods as we seek enhance and improve system performance
- Integrating UnityBL with unity_hsml